Scalability & Future Plan

Date	2 July 2026
Project Name	OptiCrop
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how OptiCrop can be scaled to handle larger user bases, increased data loads, and extended features in the future. It also captures the team's roadmap for enhancing and evolving the platform beyond its current prototype, so that it can reliably serve farmers, agricultural researchers, and policymakers at scale.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited crop and regional data coverage in the current dataset	Recommendations may be less accurate for crops or regions not well represented in the training data	High
2	No real-time IoT/sensor integration; relies on manual data entry	Users must manually enter N, P, K, temperature, humidity, pH, and rainfall values, reducing accuracy and convenience	Medium
3	No mobile or offline access	Limits usability for farmers in areas with poor internet connectivity	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Handles a limited number of concurrent users, suitable for prototype/demo scale	Deploy on a cloud platform (e.g., AWS/Azure) with auto-scaling and load balancing to support thousands of concurrent farmers, researchers, and policymakers
2	Data Storage	Uses a single local database/flat files to store crop and environmental data	Migrate to a managed, scalable cloud database with regional/crop-based partitioning to handle larger, growing datasets
3	Performance	ML model inference runs synchronously for each individual request	Introduce model caching, batch processing, and asynchronous inference pipelines to reduce response time under higher load
4	Security	Basic authentication with limited access control	Implement role-based access control, encryption at rest and in transit, and secure API gateways for different user roles (farmer, researcher, policymaker)

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Integrate real-time IoT sensor and live weather feed data	Q3 2026	More accurate, up-to-date crop recommendations with less manual data entry
Phase 2	Launch mobile app with offline mode and regional language support	Q4 2026	Wider adoption among farmers, including those in low-connectivity areas
Phase 3	Expand crop and regional database; add yield prediction analytics dashboard	Q1 2027	Broader applicability and deeper insights for researchers and policymakers
Phase 4	Integrate with government agri-schemes and marketplace linkage	Q2 2027	End-to-end value for farmers, from recommendation through to market access